

3(a)(i)	any pair (max 2) from: 1 cristae / folds, let inner membrane expand ; 2 when water enters (matrix), by osmosis / down water potential gradient ; 3 inner membrane (relatively) impermeable to water ; 4 (so) water does not enter (matrix), by osmosis / down water potential gradient ; 5 inner membrane moves (H⁺) ions out of matrix ; 6 so less water enters (matrix), by osmosis / down water potential gradient ;	2
3(a)(ii)	any three names plus explanations from: 1 <u>pyruvate</u>, for link reaction / to bind to coenzyme A / to make acetyl (CoA) / to make reduced NAD / be dehydrogenated ; 2 <u>oxaloacetate</u> to, accept acetyl / make citrate ; 3 <u>citrate</u> to, make reduced NAD / be dehydrogenated ; 4 <u>enzymes</u> to catalyse, link reaction / Krebs cycle ; 5 <u>oxygen</u> to, accept electrons / accept protons / form water ; 6 <u>water</u> as a, solvent / medium for reactions ; 7 <u>DNA</u> / <u>RNA</u>, to make (named) respiratory, enzymes / proteins ;	3

Question	Answer	Marks
3(a)(iii)	any one from: 1 so H^+, cannot move through / must move through ATP synthase ; 2 to maintain proton gradient ;	1
3(b)(i)	any four from: 1 (H atoms) split into protons and electrons ; 2 electrons, flow / move, down ETC ; 3 (releases) energy used to move H^+ to intermembrane space ; 4 more / build-up of, H^+ / positive charge, in intermembrane space ; 5 (causes / sets up) proton / electrochemical, <u>gradient</u> ;	4
3(b)(ii)	any two from: 1 (P_i) by facilitated diffusion or through a protein, channel / carrier ; 2 P_i and H^+ move together ; 3 (as) H^+ ions <u>diffuse</u> (through ATP synth(et)ase / to matrix) ;	2
3(b)(iii)	1 constant / sufficient / correct, supply / amount of, ADP / reactant ; 2 (so) ATP can continue to be made / so enough ATP can be made ;	2